

Week 1

Forces, free-body diagrams, and project goals

Automotive Aerodynamics Independent Study

Industrial Design × Physics (DePaul)

Physics mentor · 10-week IS

Today's agenda

1. Course contract: design + measurable physics
2. Translate the project proposal into force language
3. Newton review → vehicle free-body diagrams
4. Downforce, normal force, and "aero grip"
5. Drag in steady cruise
6. Modular parts: claims vs. quantities
7. Success metrics & labeling
8. Physics checkpoint (worked)
9. Deliverables for Week 1

Learning targets (by end of Week 1)

You will be able to:

- Restate the project in **measurable** physics language (forces, not vibes)
- Draw FBDs for a car: weight, normal forces, drag, downforce, drive/brake
- Explain why downforce can increase cornering grip **without** adding engine power
- Define 2–3 success metrics your scale tests can actually check

Who this course is for

You	Industrial Design; full algebra-based intro physics done
Path	Physics minor in progress
This IS	Not a substitute for calculus fluids — a bridge that <i>uses</i> forces quantitatively
Mentor role	Physics models, experiment design, honest scaling, feedback on claims

Core bet: CAD without force models fails the Physics bar; force models without a build fail the design motivation. We do both.

The student proposal (compressed)

Question: How do spoilers, wings, splitters, and diffusers change the forces on a car?

Method:

1. Research modern performance aero
2. CAD a **modular** scale vehicle (swap parts, keep body fixed)
3. Build a test platform + sensors
4. Compare configurations under controlled airflow

Our job this week: make that plan **physically precise** before printing plastic.

Forces, not vibes

Vague design talk	Physics talk (Week 1)
"More aggressive aero"	Larger downforce D_w and/or drag F_D at a stated speed
"Better grip"	Larger tire normal force $N \Rightarrow$ larger available friction
"Looks fast"	Not a Week 1 metric (save aesthetics for captions later)
"Works great"	Measured ΔF at fixed airspeed, with replicates

If you can't put it on an FBD, it's not yet a physics claim.

Learning outcome LO1 (syllabus)

Apply Newton's laws and free-body diagrams to vehicles under aerodynamic load.

Week 1 owns this outcome. Later weeks add pressure, C_D/C_L , Bernoulli limits, and Reynolds number — but **every** later claim still comes back to forces on an FBD.

Newton's laws (algebra refresh)

Law 1: If $\sum \mathbf{F} = 0$, velocity is constant (including rest).

Law 2:

$$\sum \mathbf{F} = m\mathbf{a}$$

Law 3: Forces come in pairs — useful later for tire–road interaction; today we draw **on the car**.

Today's special case: level road, constant speed $\Rightarrow \mathbf{a} = 0 \Rightarrow$ each direction balances.

Free-body diagram rules

1. Draw the system as a blob or simple side-view outline.
2. Replace the world with **force arrows** only (no “motion arrows” on the FBD).
3. Label every force with a symbol and (when known) magnitude.
4. Choose axes: x along the road, y vertical.
5. Write $\sum F_x = 0$, $\sum F_y = 0$ for steady cruise / rest.

Tip: Separate “what the air does” into components: drag (along flow) and downforce/lift (perpendicular). Airflow marks are *not* forces — they are context.

FBD 1 — Car at rest (no aero)

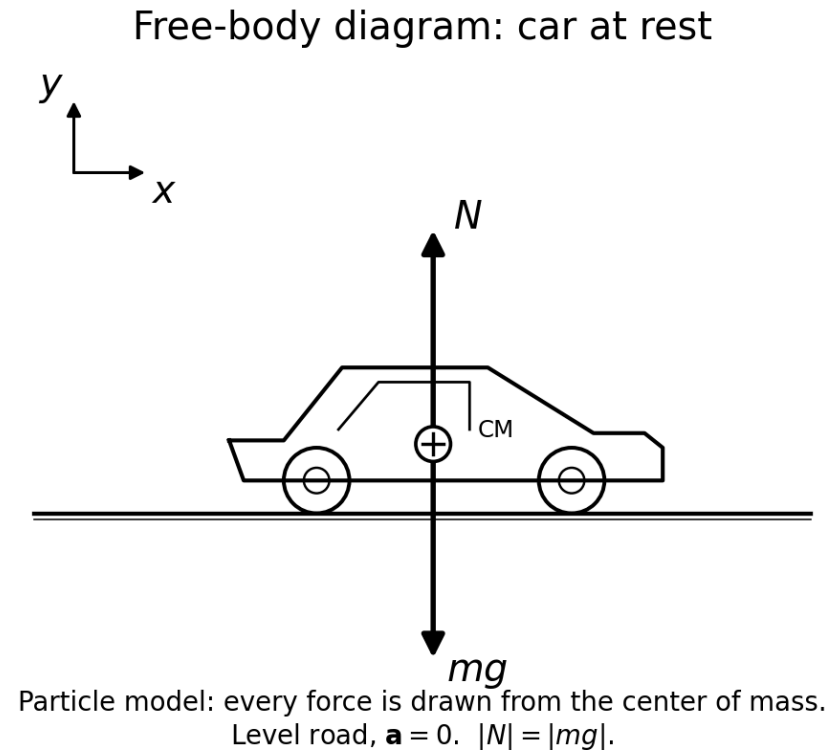
Vertical only:

$$\sum F_y = N_{\text{total}} - mg = 0$$

$$N_{\text{total}} = mg$$

- mg : weight (down)
- N : road normal (up)
- No horizontal forces needed on level ground

Language: N is not “the weight,” but it equals mg here.



Checkpoint preview — Problem 1

A 1500 kg car sits at rest. Total normal force from the ground (ignore aero)?

Use $g = 9.8 \text{ m/s}^2$:

$$N = mg = (1500)(9.8) = 1.47 \times 10^4 \text{ N}$$

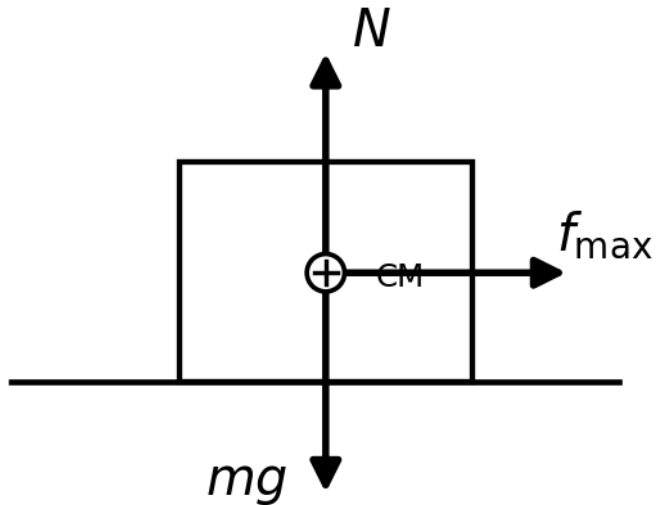
About 14.7 kN shared among the tires (distribution front/rear is a later story).

Friction needs normal force

Approximate tire limit (Coulomb-style teaching model):

$$f_{\max} \sim \mu N$$

Friction budget: $f_{\max} \sim \mu N$



Available horizontal force (same μ)

no aero

 μmg

with downforce

 $+ \mu D_w$

Cornering and braking spend this budget.
Aero raises N without changing mass m .

What is downforce?

Downforce D_w : net aerodynamic force pushing the car **into the road** (useful for grip).

In aircraft language this is **negative lift**:

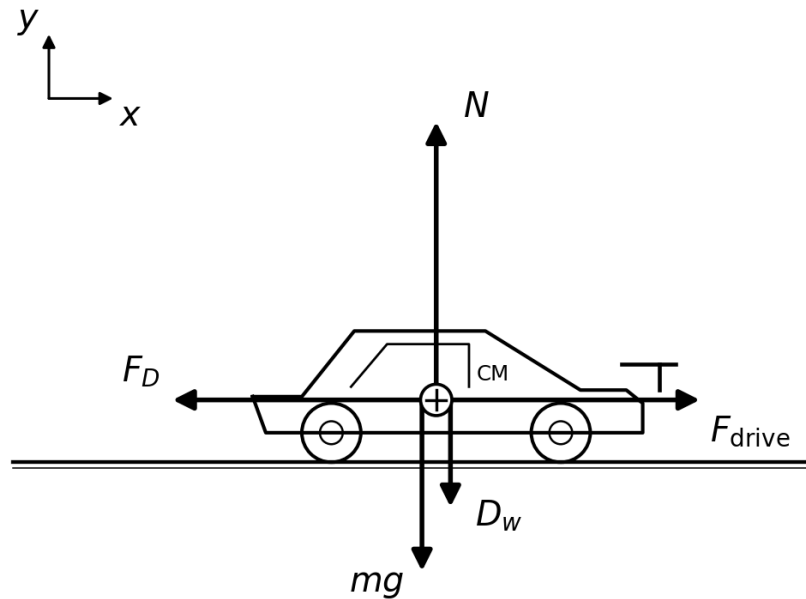
$$D_w = -F_L \quad (\text{when } F_L \text{ is upward-positive})$$

This course: we often say “downforce” in words and use $D_w > 0$ downward, or report $F_L < 0$. **Pick one convention and keep it.**

FBD 2 — Steady cruise with aero

Assume constant velocity on level road: $N = mg + D_w$ and $F_{\text{drive}} = F_D$.

Free-body diagram: steady cruise with aero



$\mathbf{a} = 0$: $N = mg + D_w$ (vertical) and $F_{\text{drive}} = F_D$ (horizontal).
Arrow length is qualitative. Airflow is not a force and is omitted.

Side-view force inventory

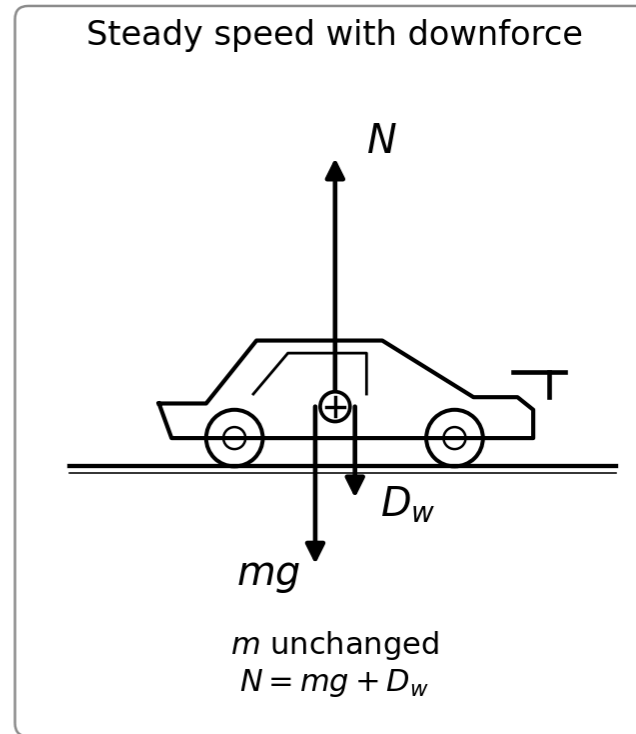
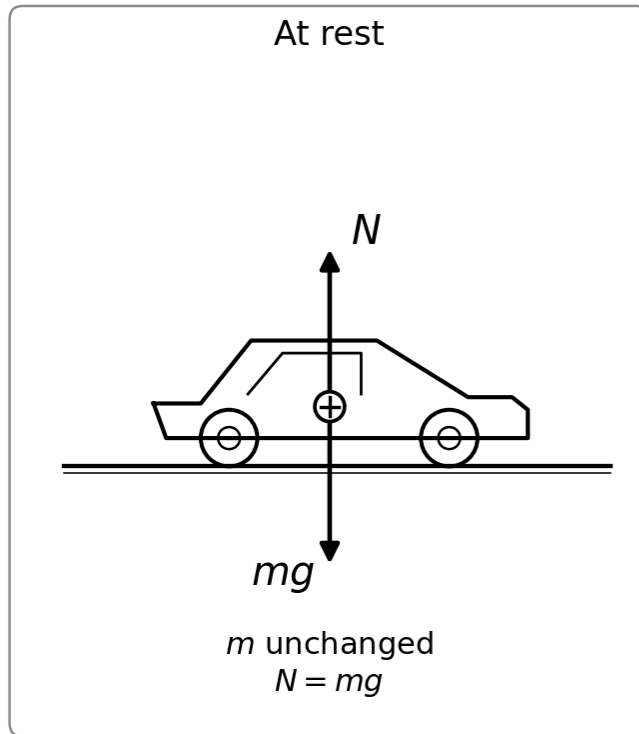
Symbol	Direction	Meaning
mg	down	Weight
N	up	Road normal (total or per axle)
D_w	down	Aero downforce
F_D	aft	Aero drag
F_{drive}	forward	Tire propulsion (cruise)
F_{brake}	aft	Tire braking (when slowing)

Optional later: lift (up), side force, pitching moments. Week 1 = nets in x and y .

The #1 misconception

"Downforce makes the car lighter."

Downforce does not make the car lighter



Say: "Aero increases load on the tires," not "aero reduces weight."

Why that matters for design

Without aero, grip budget $\sim \mu mg$.

With downforce:

$$f_{\max} \sim \mu(mg + D_w)$$

You can corner / brake harder **at speed** if D_w is large — classic race-car idea.

Tradeoff (foreshadow): Parts that make D_w often also raise F_D . More grip may cost more power to hold speed. Weeks 3 and 7 make this quantitative.

Checkpoint preview — Problem 2

Same 1500 kg car at speed: $D_w = 2000 \text{ N}$, $F_D = 800 \text{ N}$. Steady cruise.

Vertical:

$$N = mg + D_w = 14700 + 2000 = 16700 \text{ N}$$

Horizontal: drive force balances drag:

$$F_{\text{drive}} = F_D = 800 \text{ N}$$

Normal force **rose by** 2000 N (~14% in this example). Drag did **not** cancel weight; it is a separate horizontal balance.

"Aero grip" in one paragraph (checkpoint 3)

Target explanation (model answer):

Grip depends on how hard the tires are pressed into the road. At rest that press is just the car's weight. At speed, aerodynamic downforce adds an extra downward force, so the road's normal force increases even though the car's mass is unchanged. Larger normal force allows larger friction forces for cornering or braking. That is "aero grip": Newton's laws plus $f_{\max} \sim \mu N$, not magic airflow cosmetics.

Your paragraph should hit: **mass unchanged, N increases, friction can increase.**

Drag is the price of speed (and often of downforce)

In steady cruise:

$$F_{\text{drive}} = F_D$$

Power (preview, not required today):

$$P = F_{\text{drive}} v = F_D v$$

Higher drag at the same speed $\Rightarrow$ more power from the powertrain (or a lower top speed for a given power).

Industrial design implication: a spectacular wing that doubles drag may be the wrong product goal.

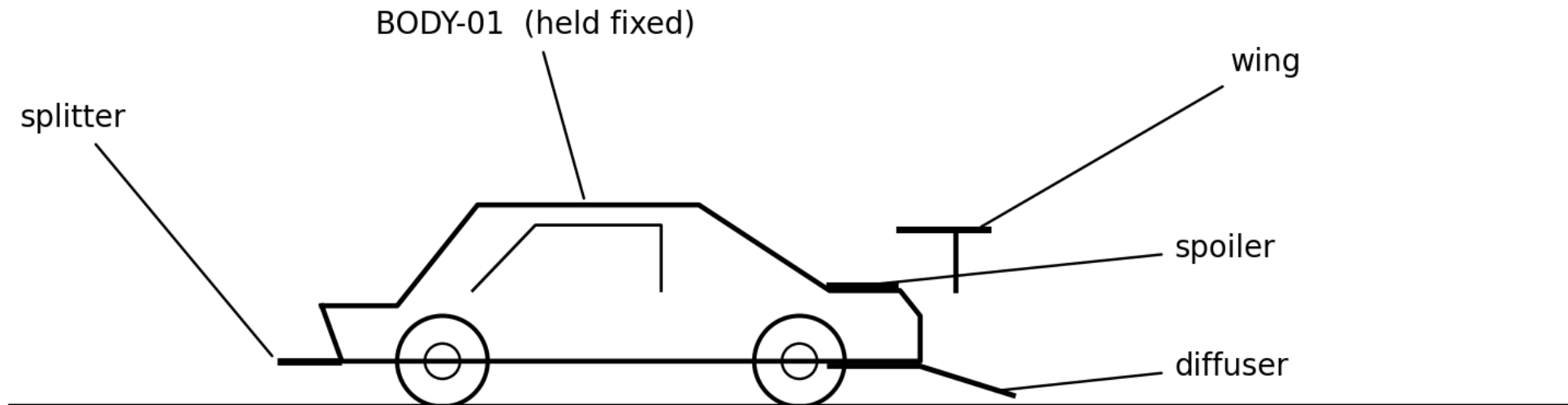
Translate design claims → physics

Design claim	Physics translation
"More grip"	Larger N and/or better $\mu \rightarrow$ larger usable tire force
"More stable at speed"	Often load distribution / moments (advanced); start with net D_w and F_D
"More efficient aero"	High D_w/F_D or meet a D_w target with less drag
"Cleaner look"	Aesthetic — fine, but separate from force metrics

Week 1 homework: write this table for **your** modular parts list.

Modular parts — what each is *claimed* to do

Modular kit: swap one part at a time



Keep the body, ride height, and alignment fixed so a force change can be attributed to one part.

Modular parts — physics quantities to watch

Part	Common claim	Watch
Spoiler	Mild downforce / wake	D_w , F_D , rear load
Wing	Strong downforce vs. angle	D_w , F_D , stall
Splitter	Front underbody / stagnation	Front N , F_D
Diffuser	Pressure recovery	Ride height, D_w , F_D

Claims are hypotheses. **Sensors** decide.

Keep the body fixed — science reason

Modular rule: change **one** aero part (or one angle) at a time.

Otherwise you cannot tell whether ΔF came from the wing, the splitter, or an accidental ride-height change.

Experimental design starts now — not in Week 6.

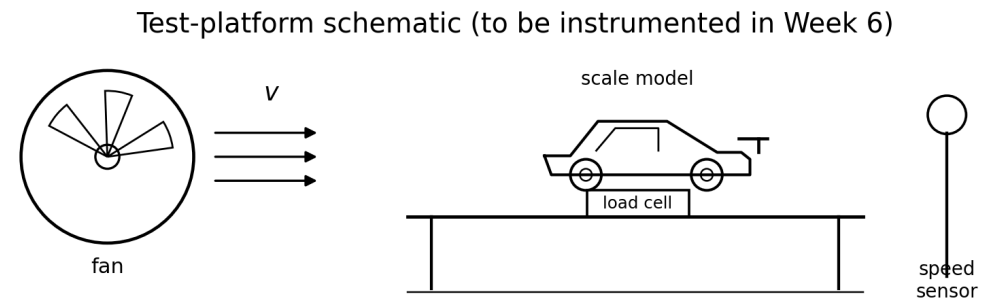
What we will measure later (so metrics make sense now)

Eventually (Weeks 6–9):

- Airspeed v (or calibrated fan setting)
- Drag force F_D
- Vertical load change related to D_w
- Geometry: AoA, ride height

Week 3 wraps these as C_D, C_L .

Week 1: **forces are the truth layer.**



Scale-model reality check

Example from the notes: 12.0 kg model

$$N_{\text{rest}} = mg \approx 118 \text{ N}$$

If aero downforce is only 2.5 N:

$$N \approx 120.5 \text{ N}$$

That's a ~2% change in total normal — easy to lose in noise if you only weigh the whole model crudely.

Implication: plan sensitive force sensing or a fixture where the load cell sees aero force clearly.

Define success metrics (due this week)

Pick 2–3 that a sensor can check. Examples:

1. At fixed fan setting, adding WING-01 increases measured downforce by $\geq X$ N vs. baseline (3 replicates).
2. Repeatability: same config, σ of F_D within $Y\%$ of the mean.
3. Efficiency interest: maximize D_w / F_D among tested wings at that speed.
4. Process: every run labeled with config ID + photo of setup.

Avoid: “looks aerodynamic,” “feels faster,” unverified full-scale top speed claims.

Labeling scheme (required)

Physical part ↔ CAD file ↔ data row:

ID pattern	Example
BODY-##	BODY-01
WING-##	WING-01
SPOIL-##	SPOIL-01
SPLIT-##	SPLIT-02
DIFF-##	DIFF-01
BASE-00	body with no aero bolt-ons

Put the same ID on the printed part (marker/engraving) and in the CSV.

One-page project brief (structure)

1. **Design question** (one sentence)
2. **Physics quantities** you will change/measure (D_w , F_D , ...)
3. **Success metrics** (2–3, measurable)
4. **Modular parts list** with claimed function each
5. **Test platform sketch** (box + fan + fixture is fine)
6. **Open risks** (sensor sensitivity, unknown airspeed, ...)

Submit as `docs/week-01-project-brief.md` (or PDF) in the repo.

Worked checkpoint — full solutions

1. $N = mg = 1500 \times 9.8 = 1.47 \times 10^4 \text{ N}$.
2. Cruise FBD: $N = 16700 \text{ N}$; $F_{\text{drive}} = 800 \text{ N}$ balances drag. Downforce raised N ; drag is horizontal.
3. Aero grip paragraph: mass fixed; $N = mg + D_w$; $f_{\text{max}} \sim \mu N$.

Bring your own wording to discussion — mentor will watch for “lighter car” language.

In-class sketch exercise (10 minutes)

Cover the next slides. Draw **two** FBDs side by side:

A. Car at rest B. Steady cruise with D_w and F_D

Checklist:

- ☐ Weight down
- ☐ Normal up
- ☐ Downforce down (on B)
- ☐ Drag aft (on B)
- ☐ Drive forward (on B)
- ☐ No “velocity arrow” pretending to be a force

Then compare to the rest and cruise figures.

Design / build tasks this week

- [] List modular parts + one-sentence **claim** each
- [] Define 2–3 success metrics
- [] Sketch or photograph intended test platform
- [] Choose labeling scheme and start a BOM stub in `cad/bom.md`
- [] Optional: mood-board of reference cars — but caption each with a *force hypothesis*

Physics before filament when possible.

Deliverables checklist

Deliverable	Where
Checkpoint answers (1–3)	notes or curriculum/week-01-checkpoint.md
One-page project brief	docs/week-01-project-brief.md
Parts list + IDs	brief or cad/bom.md
Platform sketch/photo	docs/ or cad/notes/

Due before Week 2 meeting.

How you will be assessed on Week 1

From the syllabus spine: weekly checkpoints matter (suggested 30% of course).

Strong Week 1 work:

- Correct FBDs and $N = mg + D_w$
- Clear metrics
- No “downforce = lighter”

Weak Week 1 work:

- Only aesthetic renders
- Metrics you cannot measure
- Aero claims with no force symbols

Looking ahead — Week 2

Pressure, density, and dynamic pressure

$$q = \frac{1}{2} \rho v^2$$

Why aero forces grow roughly like v^2 , and how to sketch high/low pressure faces on a wing or splitter.

Bring your parts list — we will annotate pressure hypotheses on your sketches.

Instructor talking points (optional slide)

- Correct “lighter” language early; it resurfaces all term.
- If the student jumps to CAD: ask “what force changes, by how much, at what speed?”
- Sensor sensitivity: surface the 2% example so Week 6 calibration is motivated.
- Keep Week 1 algebra-only; save Bernoulli fights for Week 4.

References for this lecture

- Course note: [physics/01-forces-and-fbds.md](#)
- Week plan: [curriculum/week-01-forces-and-goals.md](#)
- Proposal: [docs/proposal.md](#)
- Syllabus LO1: [curriculum/SYLLABUS.md](#)
- Figures: [curriculum/slides/figures/](#) (original course diagrams)

End of Week 1 lecture

Next actions for the student

1. Finish checkpoint in your own words
2. Write the one-page brief
3. Sketch the platform
4. Push to the course GitHub repo

Questions?